THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A metal oxide X dissolves in water to turn red litmus blue. X is most likely: 1

(a) Fe_2O_3 **(b)** Na_2O **(c)** CuO **(d)** Al_2O_3

2. A metal of atomic number 11 reacts with water. The solution formed: 1

(a) turns litmus colourless **(b)** turns red litmus blue
(c) is neutral **(d)** turns blue litmus red

3. Assertion (A): Zinc is more reactive than iron, so galvanised iron does not rust even when scratched. Reason (R): Zinc corrodes in preference to iron. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A metal M forms M_2O_3 (an amphoteric oxide). Its chloride formula is: 1

(a) MCl_3 **(b)** MCl_2 **(c)** M_2Cl_3 **(d)** MCl

5. The metal which can be extracted by electrolysis of its molten oxide in presence of cryolite is: 1

(a) iron **(b)** copper **(c)** zinc **(d)** aluminium

6. Assertion (A): Copper displaces silver from silver nitrate solution. Reason (R): Copper is more reactive than silver. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. In electrolytic refining of copper, 0.5 mol of Cu^{2+} ions are reduced at the cathode. The mass of copper deposited (Cu = 63.5) is: 1

(a) 63.5 g **(b)** 15.9 g **(c)** 127 g **(d)** 31.75 g

8. An alloy used for electrical fuse wires has low melting point. It consists of: 1

(a) iron and carbon **(b)** tin and lead **(c)** copper and tin **(d)** copper and zinc

9. Assertion (A): Carbon is used to reduce zinc oxide but not sodium oxide. Reason (R): Sodium is more reactive than carbon. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. In the thermit reaction, 54 g aluminium reacts with excess Fe_2O_3 (Al = 27, Fe = 56). The mass of Fe produced is: 1

(a) 112 g **(b)** 168 g **(c)** 56 g **(d)** 54 g

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does silver not react with dilute HCl but zinc does? Which metal could displace hydrogen from water? 2

12. An ionic compound has a high melting point but is brittle. Explain. 2

13. Why do metals not react with dilute HNO_3 to give hydrogen? 2

14. Why do we not use zinc to coat food cans? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the surface of copper vessels turns green and how it can be cleaned using lemon or tamarind. 3

16. Explain the steps to extract copper from copper pyrites in short, with the final purification. 3

17. Explain how brass is better than copper for making statues and utensils. 3

OR

Metal X is stored under kerosene and forms X_2O, which dissolves in water forming a strong base. Identify X and write equations. 3

18. Why are silver and gold found in a free state while sodium is found as compounds? 3

OR

Name three uses of aluminium and relate each to a property. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Calculate the mass of zinc obtained from 162 g of zinc oxide using carbon (Zn = 65, O = 16, C = 12) and the volume of CO at STP. 5

OR

Explain electrolytic reduction of sodium chloride and electrolytic refining of copper and compare the two processes. 5

20. An element X (atomic number 20) reacts with chlorine (atomic number 17). Write electronic arrangements, show the formation of the compound, formula and four properties. 5

OR

Differentiate between: ore and mineral, roasting and calcination, alloy and amalgam, corrosion and galvanisation, ionic and covalent compounds (one point each). 5

END OF QUESTION PAPER — **Total: 40 Marks**